

Subject Description Form

Subject Code	AAE5202
Subject Title	Advanced Aircraft Structures and Materials
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: ME577 Advanced Aircraft Structures
Objectives	1. To provide students an overview of the structures in modern aircraft. 2. To provide students with tools that are needed to formulate and solve problems concerning compression/tension, bending, torsion and buckling in aircraft structures. 3. To provide students with an overview of the advanced materials that are used for aircraft vehicles. 4. To provide students with an overview of the non-destructive testing techniques that are used to ensure the safe operation of aircraft vehicles.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. perform stress analysis for typical aircraft structural components using both analytical methods and computational tools; b. obtain in-depth understanding of the mechanical behavior of the materials that are used for aircraft vehicles; c. choose the non-destructive testing methods that best suit certain aerospace structural components; and d. recognize the frontier of research in aircraft structures and materials.
Subject Synopsis/ Indicative Syllabus	Structures: Fuselage; Wing; Tail; Landing gear; Thin-wall beams; Tapered beams; Ribs; Cut-outs; Loads applied on airframes; Stress analysis of aircraft structural components Materials: Typical aircraft materials and material characteristics; Characteristics of composite materials Non-destructive testing and evaluation of aircraft structures (NDT&E): Finite element method (FEM) for the analysis of aircraft structures

Teaching/Learning Methodology	Lectures, tutorials and guided study by project/case study/literature survey are used to deliver the fundamental knowledge and research elements in relation to aircraft structures and materials.																																	
	Teaching/Learning Methodology		Outcomes																															
			a	b	c	d																												
	Lecture		√	√	√	√																												
	Tutorial/Guided Study		√	√	√	√																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>2. Take-home assignments and/or in-class quizzes</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Final examination</td><td>60%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.4 × Continuous Assessment + 0.6 × Final Examination The assignments and quizzes are used to assess the students' understanding of the stress analysis methods and their capabilities of mathematical problem formulation and programme application for typical aircraft structures. The final examination will be conducted to evaluate the students' performance in all the topics of the syllabus with a limited examination time.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	2. Take-home assignments and/or in-class quizzes	40%	√	√	√	√	3. Final examination	60%	√	√	√	√	Total	100%				
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3. Final examination	60%	√	√	√	√																													
Total	100%																																	
Student Study Effort Expected	Class contact:																																	
	▪ Lecture					39 Hrs.																												
	Other student study effort:																																	
	▪ Self-learning					45 Hrs.																												
	▪ Project report preparation					22 Hrs.																												
	Total student study effort					106 Hrs.																												

Reading List and References	1. Sun C. T., Mechanics of Aircraft Structures, John Wiley & Sons, latest edition. 2. Megson, T. H. G., Aircraft Structures for Engineering Students, Elsevier, latest edition. 3. Gibson, R. F., Principles of Composite Material Mechanics, McGraw-Hill, International Editions, latest edition.
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Jun 2024